

Our polylogarithm ladders as Bloch-group elements: the Zagier K -theory picture

Vladimir Reshetnikov

2026-06-21

Research report. Created (UTC) 2026-06-21T21:41:24Z; repository HEAD
ce20ea60a460c71d5f24caf4d2551e867036bf69. Store zagier-bloch-framework.wl.
Companion to golden-polylog-ladders; reads Zagier’s paper in
docs/pdfs/PolylogsDedekindZetaAndKTheory/.

Abstract

The golden, minimal-Pisot, and supergolden polylogarithm ladders we computed are not isolated curiosities: each is an element of a *Bloch group* $\mathcal{B}_m(F)$, and in Don Zagier’s framework relating polylogarithms, Dedekind zeta values, and algebraic K -theory, each maps under the *modified polylogarithm* P_m to a rational multiple of $\zeta(m)$. We make the identification explicit, observe that our minimal-Pisot base ω is Zagier’s flagship “spectacular” ladder example (the field $\mathbb{Q}(\theta)$, $\theta^3 - \theta - 1 = 0$, discriminant -23), and—most usefully—resolve the open question from the base hunt: for the complex cubic fields (ω , supergolden), real-argument ladders exist at every weight but give only $\zeta(\text{even})$ at even weights, while the genuine field invariant $\zeta_F(\text{even})$ is carried by the *imaginary* part of P_m at the complex embedding. This is why a real-argument weight-4 ladder search returns “none”—the content is not real.

1 The modified polylogarithm

Zagier’s one-valued modified polylogarithm (his eq. (33)) is

$$P_m(x) = \operatorname{Re}_m \left(\sum_{j=0}^m \frac{2^j B_j}{j!} (\log |x|)^j \operatorname{Li}_{m-j}(x) \right), \quad \operatorname{Re}_m = \begin{cases} \operatorname{Re}, & m \text{ odd}, \\ \operatorname{Im}, & m \text{ even}, \end{cases} \quad (1)$$

with B_j the Bernoulli numbers and $\operatorname{Li}_1(x) = -\log(1-x)$. It is continuous on the Riemann sphere away from $\{0, 1, \infty\}$, satisfies $P_m(\bar{x}) = (-1)^{m-1} P_m(x)$, and $P_m(1) = \zeta(m)$ for odd m , 0 for even m ; $P_2(i)$ is Catalan’s constant. We verified our implementation against Zagier’s worked examples $P_5(-\frac{1}{8}) - 126P_5(\frac{1}{2}) - 162P_5(-\frac{1}{2}) = \frac{403}{16}\zeta(5)$ (his eq. (825)) and the long thirteen-term relation (825)/(833), to 60 digits.

2 Ladders are Bloch-group elements

Zagier’s “Ladders” section (§9C) records exactly our construction. A number u satisfying a multiplicative relation $\prod_r (1 - u^r)^{a_r} = \pm u^N$ —our *pure-unit seeds* $1 - \rho^a = \rho^b$ are the special case—yields, for each $m \geq 2$, a Bloch-group element $\xi_m = \sum_r \frac{a_r}{r^{m-1}} [u^r] \in \ker \beta_m$. Under the regulator P_m ,

the messy elementary “tower” (powers of $\log \rho$ times π^{2k} and ζ) is absorbed and a clean ζ -multiple remains. For our ladders:

$$\text{golden : } 15 P_3(\phi^{-2}) = 12 \zeta(3), \quad (2)$$

$$\text{supergolden : } -60 P_3(\rho) + 54 P_3(\rho^2) - 32 P_3(\rho^3) - 3 P_3(\rho^6) = -36 \zeta(3), \quad (3)$$

both exact. Compare (2) with the “raw” form $15 \text{Li}_3(\phi^{-2}) = 10 \ln^3 \phi - 2\pi^2 \ln \phi + 12\zeta(3)$: the modified polylogarithm is simply the natural coordinate in which a ladder *is* its ζ -value. The higher golden ladders (weights 5 through 9 in the companion report) become, in the same way, the stated $\zeta(5), \zeta(6), \zeta(7), \zeta(9)$ multiples.

3 The bases as number fields

base ρ	minimal polynomial of $1/\rho$	field F	signature	disc
golden ϕ^{-1}	$x^2 - x - 1$	$\mathbb{Q}(\sqrt{5})$	(2, 0) totally real	5
minimal Pisot ω	$x^3 - x - 1$	$\mathbb{Q}(\theta)$	(1, 1)	-23
supergolden recip.	$x^3 - x^2 - 1$	$\mathbb{Q}(\sigma)$	(1, 1)	-31

The middle row is Zagier’s own running example (§4–5, §9C): the field $\mathbb{Q}(\theta)$, $\theta^3 - \theta - 1 = 0$, whose ladder he calls “particularly spectacular,” and whose defining unit relations $1 - \theta = -\theta^{-4}$, $1 + \theta = \theta^3, \dots$ are precisely the seeds we rediscovered. Our supergolden base gives a *different* complex cubic field, discriminant -31.

4 Resolution of the even/odd question

By Borel’s theorem the rank of $K_{2m-1}(F)$ is $n_+ = r_1 + r_2$ for odd m and $n_- = r_2$ for even m , and the regulator covolume is a rational multiple of $\sqrt{\Delta} \zeta_F(m) / \pi^{mn_+}$. For a complex cubic field ($n_+ = 2, n_- = 1$) this dictates the behaviour we observed:

- **Odd weights** ($m = 3, 5, \dots$): P_m is the *real* part, $n_+ = 2$, and genuine *real-argument* ladders exist (we found the weight-3 ones for ω , supergolden, and $x^4 + x - 1$).
- **Even weights** ($m = 4, \dots$): P_m is the *imaginary* part and *vanishes on real arguments*. A real-argument ladder can only produce $\zeta(\text{even}) = \text{rational} \cdot \pi^{\text{even}}$ (the trivial $\mathbb{Q}$ -part). The field invariant $\zeta_F(\text{even})$ is carried by P_m at the *complex* embedding.

Concretely, for the supergolden base, P_4 at the complex root of $x^3 + x - 1$ equals $-0.91599 \dots \neq 0$ (the K_7 regulator), whereas $P_4(\text{real } \rho) = 0$. *This is the resolution of the open question*: the weight-4 real-argument search returns “none” not because the field is silent at weight 4, but because its weight-4 content is imaginary.

Remark 1. Whether the supergolden base climbs to weight 5 in the *real* ladder is therefore a question about how many relations $\prod (1 - \rho^r)^{a_r} = \pm \rho^N$ the specific number ρ satisfies (Zagier: the conditions to climb alternate n_- and n_+), not one that a PSLQ “no relation” can settle—consistent with the methodological caveat of the companion report. The minimal Pisot ω satisfies many such relations (Zagier/Lewin list seven beyond the defining ones) and so climbs high; whether the supergolden σ does is open.

5 The reliable climbing method, and how high each base goes

Two companion papers supply the *deterministic* method that blind PSLQ could not: Zagier’s *Bloch–Wigner–Ramakrishnan* function D_m (Math. Ann. 1990) and Cohen–Lewin–Zagier’s *A Sixteenth-Order Polylogarithm Ladder*. The recipe: (i) find the *cyclotomic relations* $\prod_n (\alpha^n - 1)^{c_n} = \pm \alpha^N$, equivalently the k for which $\Phi_k(\alpha)$ is a unit ($|\text{Norm}| = 1$), giving $I(1) = (\#\text{units}) - \text{rank } \mathcal{O}_F^\times$ independent relations; (ii) *pseudointegrate*—the m -th rung is $\sum_n \frac{c_n}{n^{m-1}} P_m(\alpha^n) = 0$, the relations living in a lattice in $\mathbb{R}^{d(m)}$ with $d(m) = r_2$ (m even), $r_1 + r_2$ (m odd; halved if α is self-reciprocal). One climbs while $I(m) = I(1) - \sum_{j=2}^m d(j) > 0$. *The maximal weight is therefore a count, not a search*, and a PSLQ “no relation” is a false negative whenever $I(m) > 0$.

Computing the cyclotomic units (via $\text{Norm } \Phi_k(\rho) = \text{Res}(p, \Phi_k)$) gives:

base	$I(1)$	signature	climbs to weight
golden $x^2 + x - 1$	2	(2, 0)	2 free; 3, 5, 7, 9 by ζ -exhaustion
supergolden $x^3 + x - 1$	4	(1, 1)	3 (then $I(4) = 0$, stops)
$x^4 + x - 1$	7	(2, 1)	4 (the weight-4 rung is imaginary)
minimal Pisot $x^3 + x^2 - 1$	8	(1, 1)	6 (8–9 with ratio-units)
Lehmer $x^{10} + x^9 - \dots + x + 1$	71	(2, 4)	16

This *settles the open question*: the supergolden base reaches weight 3 and stops ($I(4) = 0$)—its $I(1) = 4$ is simply too small—so there is no weight-5 real ladder, deterministically and not by a flaky PSLQ negative. It also corrects two earlier false negatives: the minimal Pisot ω climbs to weight 6 (not 4), and **Lehmer’s number**—declared earlier “**not a ladder base at all**”—**is in fact the champion, weight 16**. We reproduced Cohen–Lewin–Zagier’s Lehmer data exactly: the 52 cyclotomic units at $k \leq 120$ are $k = 1, 2, 3, 5, 6, 7, 8, 9, 10, 11, 13, 16, \dots, 118$, matching their list term for term.

Verification and reproducibility

P_m (1) and the identities (2)–(3), the validation against Zagier’s eqs. (825)/(833), and the complex-embedding computation $P_4(\sigma_{\mathbb{C}}) = -0.916\dots$ are in `src/PolyLog/tools/scratch/zagier/` and stored in `src/PolyLog/identities/zagier-bloch-framework.wl`.